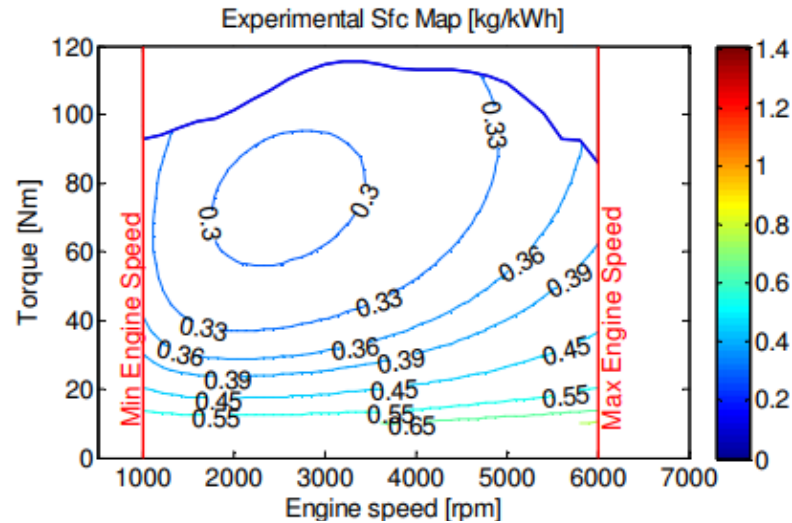


MOTORI A
COMBUSTIONE
INTERNA - 2

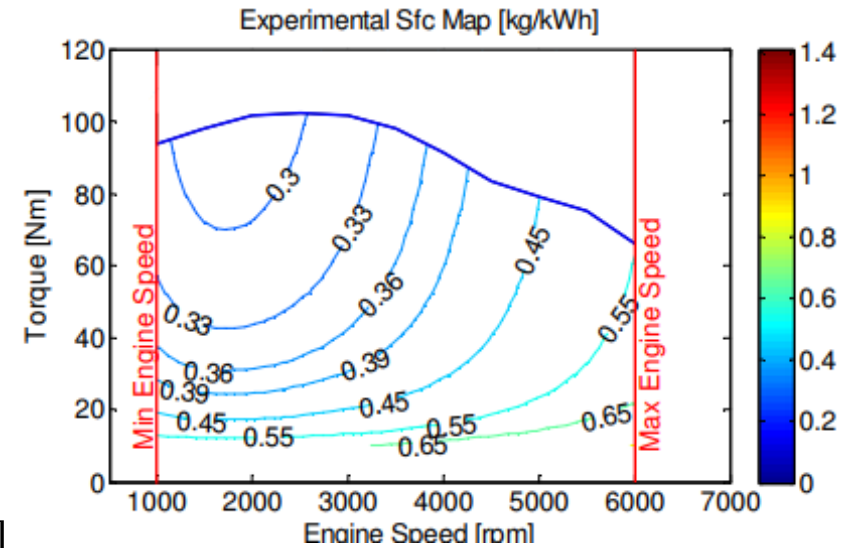


BSFC Brake specific fuel consumption

1.4 fire 8V 75CV –
SI – inline4 –
naturally aspirated

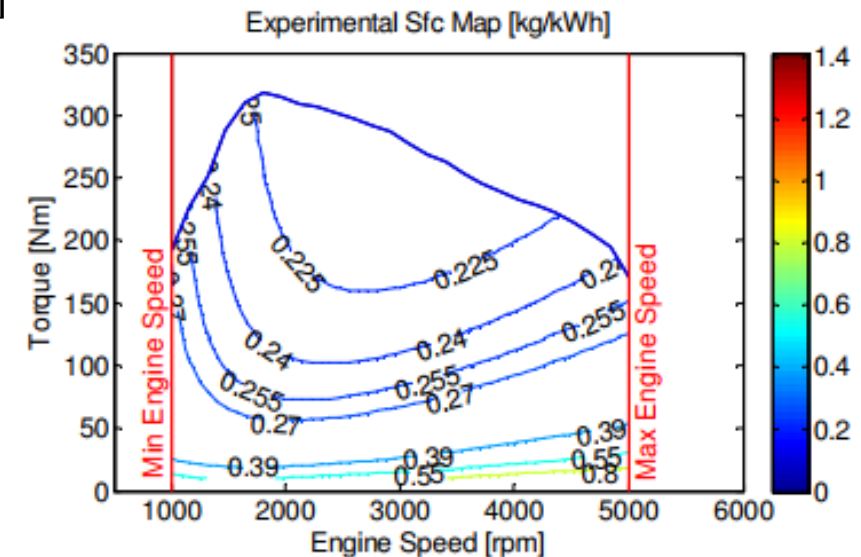
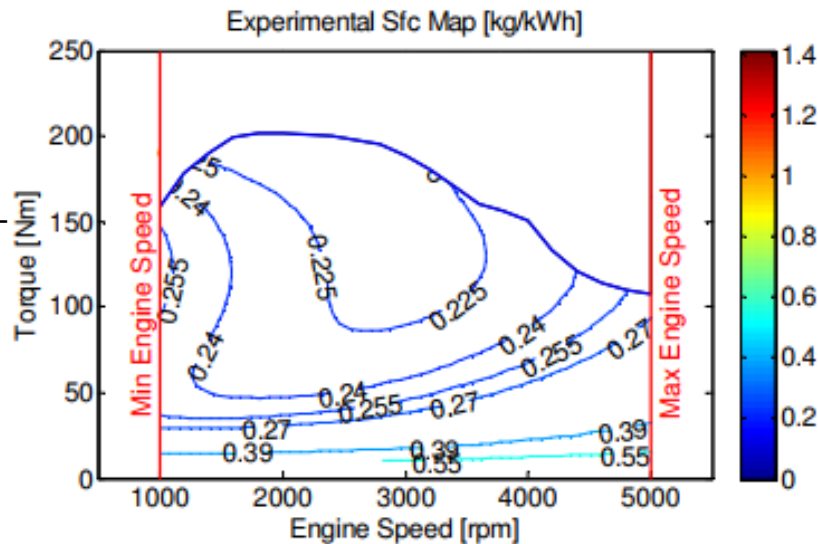


$$SFC = \left[\frac{kg}{kWh} \right]$$



1.2 fire 8V – 55CV
– SI – inline4 –
naturally aspirated

1.3 multijet 95CV –
diesel -
turbocharged



1.9 multijet 140CV
– diesel -
turbocharged

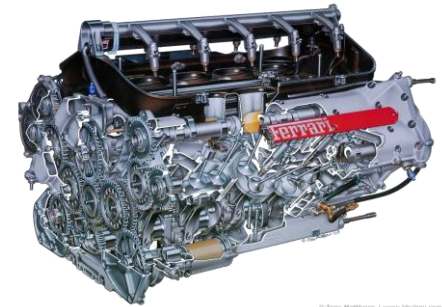
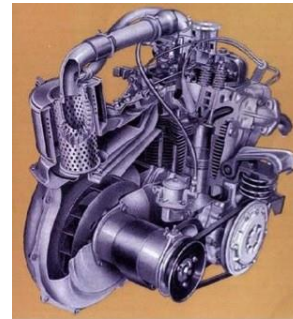
eton



Introduction

How to increase the power of an engine?

- Increase the displacement (larger cylinders or more cylinders)



- Increase the maximum engine speed

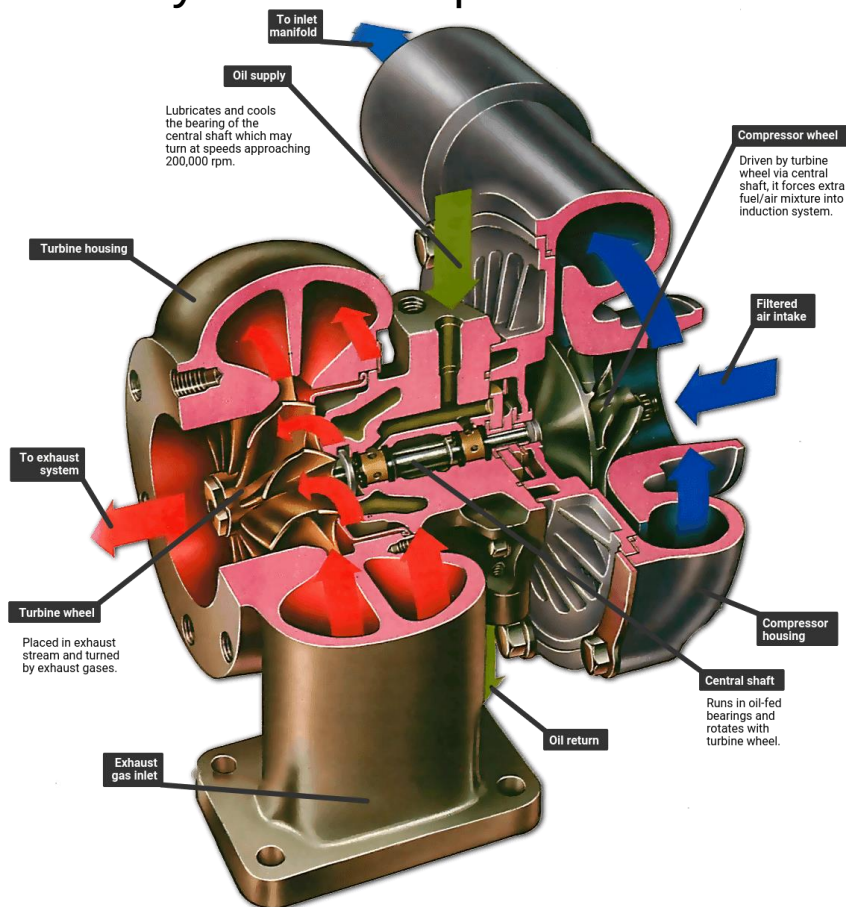


- Increase the maximum torque by increasing in-cylinder maximum pressure



Turbocharging

- Goal: More air, more oxygen, more fuel can be burned, more power can be obtained
- Practically: use a compressor that increases the pressure in the intake manifold



The compressor is mechanically connected on the same axis to **a turbine** that converts the waste energy in the exhaust gases (hot and at relatively high pressure) into mechanical power.

The mechanical power to move the compressor comes (*almost*) for free!

The turbine and compressor assembly form the **so called turbocharger**



Pros&Cons

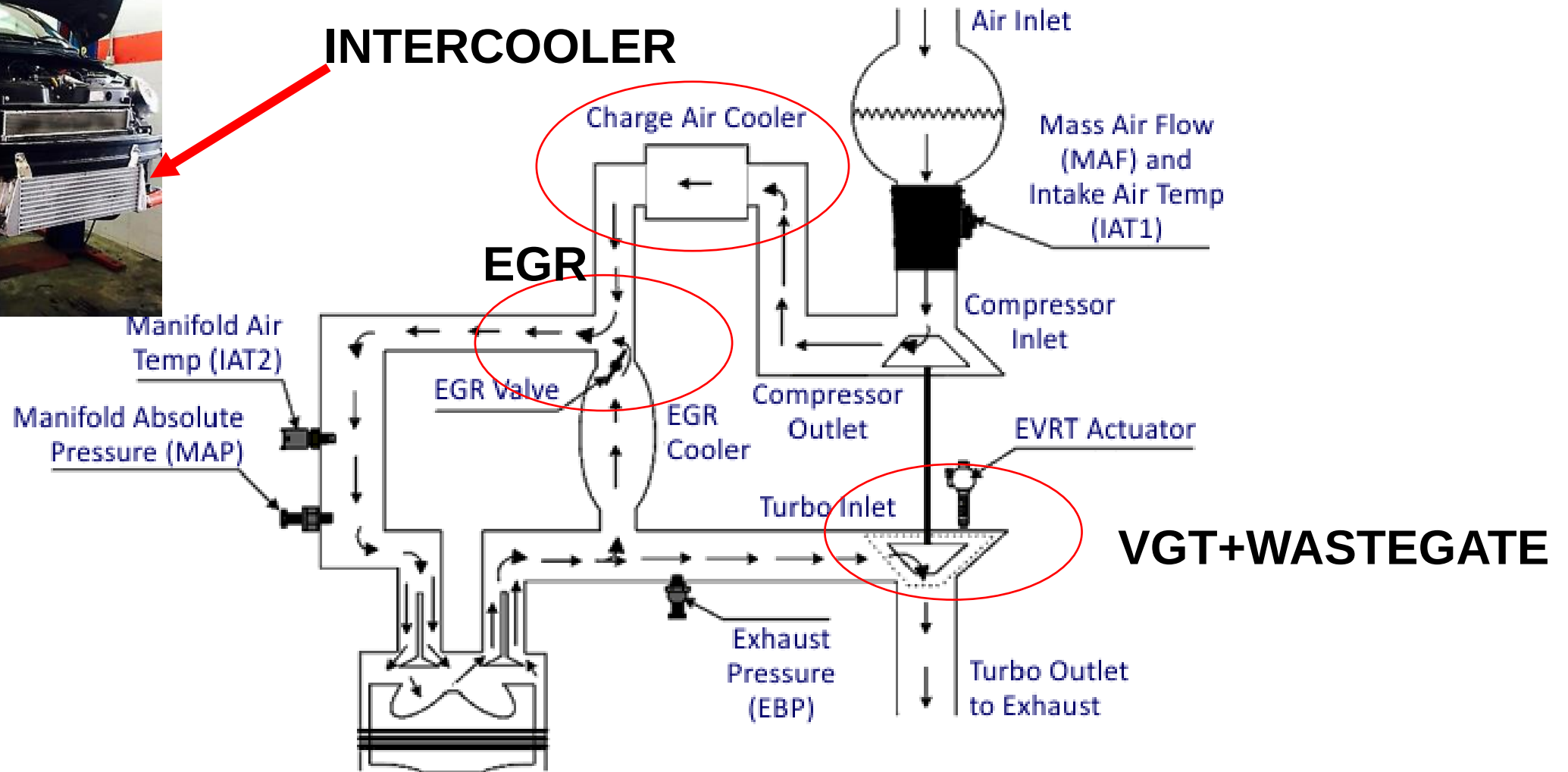
PROS

- In a SI, indirect injection engine, the main objective and advantage is to increase the torque, thus power using a smaller engine.
- The presence of the compressor that “pushes” the air into the engine, reduces the overall pumping losses (it’s the compressor that pushes the air into the cylinder and not the piston that has to “suck” the air in). This is true only if the compressor is “pumping” enough (medium-high load depending on the turbocharger).
- In a diesel engine peak power and overall efficiency are **ALWAYS** increased, just because i am using energy **that would otherwise be wasted**.
- This is true also for a SI engine without direct injection if the engine and turbocharger are matched in order that the best efficiency of the turbocharger (on the BFSC chart) is HIGH where the engine efficiency is low.

CONS

- The overall system complexity (hence cost) is increased.
- The turbocharger introduces a series of degrees of freedom in the system that have to be controlled in order to guarantee best efficiency and integrity.
- The air compressed by the compressor **may need to be cooled through and INTERCOOLER, otherwise the cycle low temperature raises and efficiency decreases.**
- **Turbine and compressor inertia may introduce a delay in the system known as turbo-lag.**

Turbocharging Overview





Turbocharging

Wastegate&Variable Geometry Turbine (VGT)



The wastegate (**WG**) valve bypasses a fraction of the hot exhausts into the exhaust line without letting it pass into the turbine. The wastegate valve opening is controlled by the ECM in order to **limit turbine speed** (the turbine revs up to 250.000rpm) and/or boost pressure **and thus the air mass pumped to the intake by the compressor**.

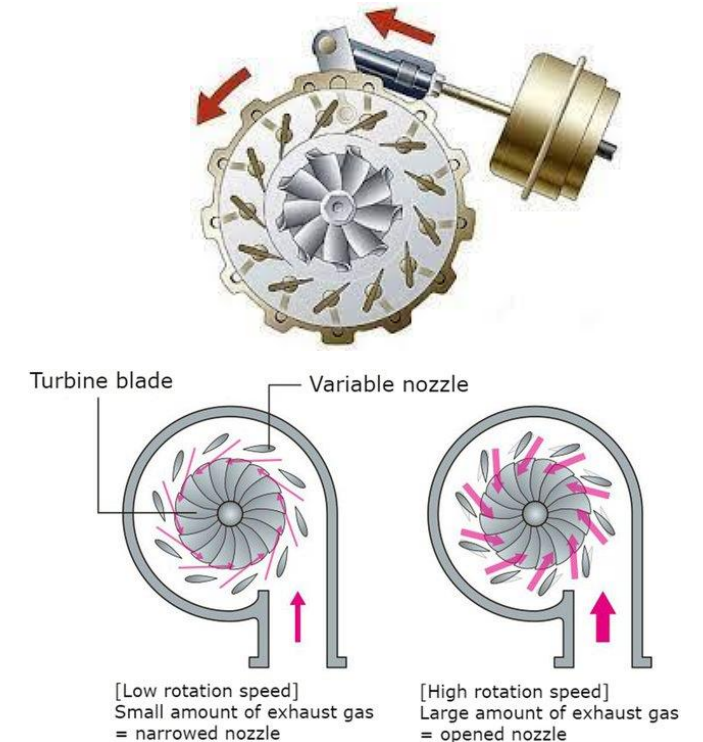
In order to do so, ECM uses information from the **EBP** (exhaust boost pressure) and **MAP** (manifold absolute pressure) **sensor**

The **variable geometry turbine (VGT)** can modify the flow conditions by adjusting the vanes through an electronic (actuator controlled by ECM) or hydraulic mechanism (engine oil, valves managed by ECM).

When the vanes are opened, the effect is the same as the wastegate (that is no longer required).

This solution allows to optimize the compressor boost pressure along the whole rpm and torque range.

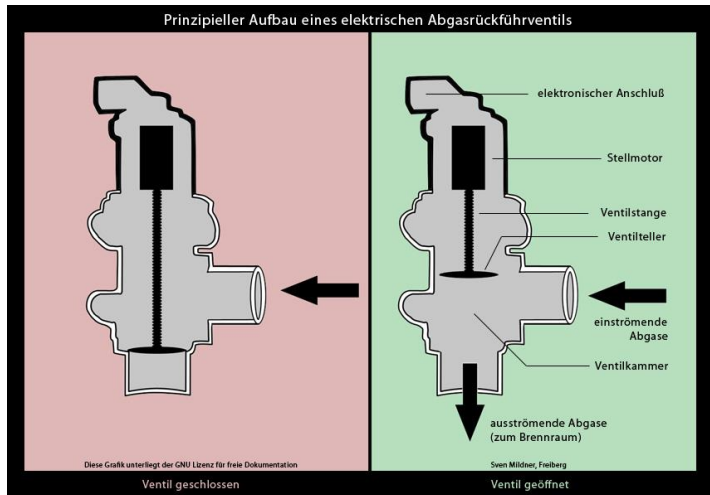
This solution gives more control on the airflow and supports control policies to reduce pollutants emission. The VGT strongly mitigates or eliminates turbo lag.





Exhaust Gas Recirculation valve

Overview



The EGR valve (controlled by the ECM), recirculates a fraction of the exhaust gases towards the intake.

The exhaust gases are **inert** in the combustion chamber. As a consequence, **this reduces the in-chamber temperature**, reducing the NOx production.

The NOx production also decreases because of the reduced oxygen availability.

With accurate engine design and calibration the soot that is recirculated in the EGR flow can be burned (smaller particles) or can be increased in size (good for the pollution, bad for the engine).

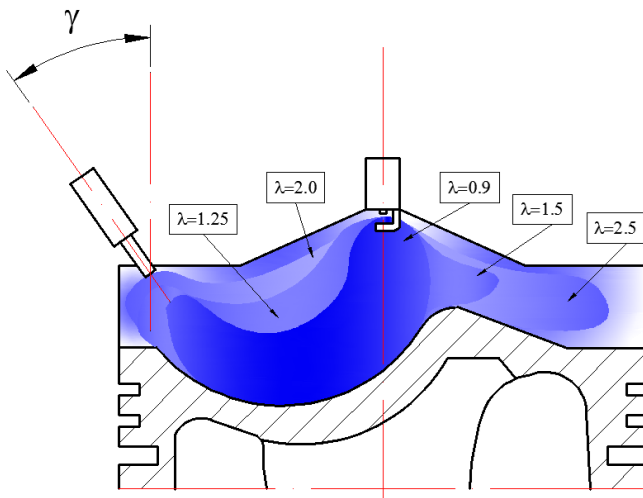
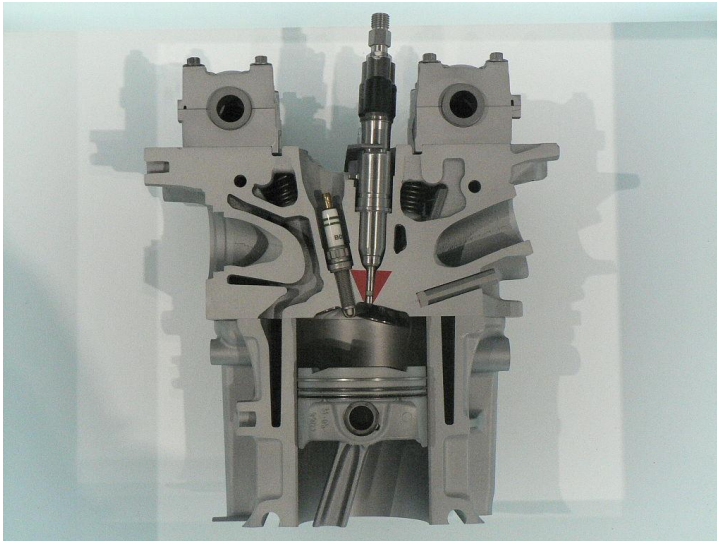
Large soot particles can be trapped more easily in the DPF but increase fouling of the exhaust lines, intake manifold, EGR valve, EGR cooler etc.

EGR flow needs to be cooled down before being forwarded to the intake manifold, this is done through a dedicated heat exchanger: **the EGR cooler**



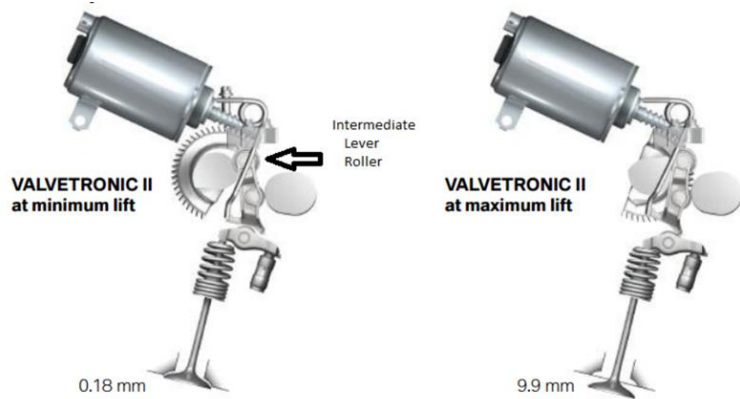


GDI

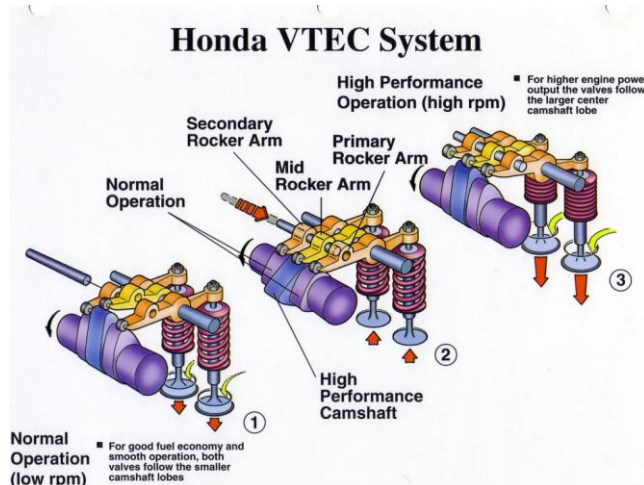


- Dishomogeneous charge (stratified) at low loads: stoichiometric (or slightly rich) mixture ($AFR=AFR_{st}$) nearby the spark plug in order to start combustion, lean to very lean mixture ($AFR \gg AFR_{st}$) away from the spark plug.
- Higher compression ratios can be achieved.
- The throttle is still present but can be kept fully open during stratified charge operation.
- At medium/high loads the ECM normally switches to homogeneous charge mode ($AFR=AFR_{st}$ everywhere, similarly to port injection).
- In-chamber fluid dynamics requires accurate design in order to obtain the **desired fuel concentration** in stratified charge mode and **adequate air-fuel mixing** in homogeneous charge mode.
- Often, a traditional port injector is installed for homogeneous charge operation.
- Critical combustion control during transients (normally switches to homogeneous). Limited advantages in case of conventional powertrains, but more potential in case of hybrid powertrains.
- Increased system complexity.
- More sophisticated fuel injectors.
- Due to limited time for air-fuel mixing, soot formation can occur.
- Due to lean combustion conditions, higher NOx emissions.

Variable Valve Lift and Timing (VVL-VVT)

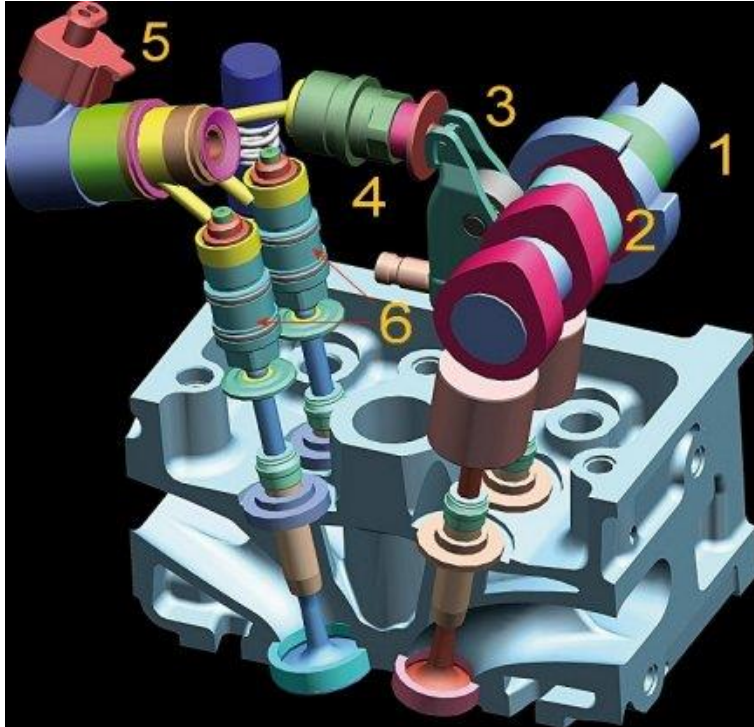


- An hydraulic , electrical or mechanical system can be adopted to vary the valve lift.
- With variable valve lift, the intake air flow can be regulated through the intake valve/s, potentially allowing the elimination of the throttle body (reduction of the pumping losses).
- Variable valve timing allows to optimize the opening and closing time of the valves, allowing a better filling of the cylinder (exploitation of dynamic effects in the manifold).
- VVT allows to obtain a more favourable torque profile (improved torque at both high and low rpm) that would not be possible with a fixed cam timing.
- Some degree of «inner EGR» can be achieved by acting on the valve timing (exhaust gases can be sucked back into the cylinder) without using an EGR loop.
- The pumping losses can be reduced by anticipating intake valve closing and exhaust valve opening.
- Almost all manufacturers have their proprietary VVL/VVT systems, often associated with other systems aimed at increasing engine efficiency and/or performance (BMW Valvetronic, Porsche: Variocam, FCA: Multiair, Honda:VTEC...).
- There are systems with continuously variable valve timing (Valvetronic, top image) and finite variable valve timing (Honda VTEC- 2 timings).





FCA Multiair



The system is hydraulic and uses the engine oil as operating fluid.

The camshaft 2 is the same for the intake and exhaust valves.

The exhaust valves are directly actuated by the traditional cammes.

The cam 2 acts on the hydraulic pump 4 (works like a syringe) through the rocker 3.

The intake valves are actuated by the hydraulic pistons 6 that receive the pressurized oil from the pump 4 **through the electronically controlled electrovalve 5.**

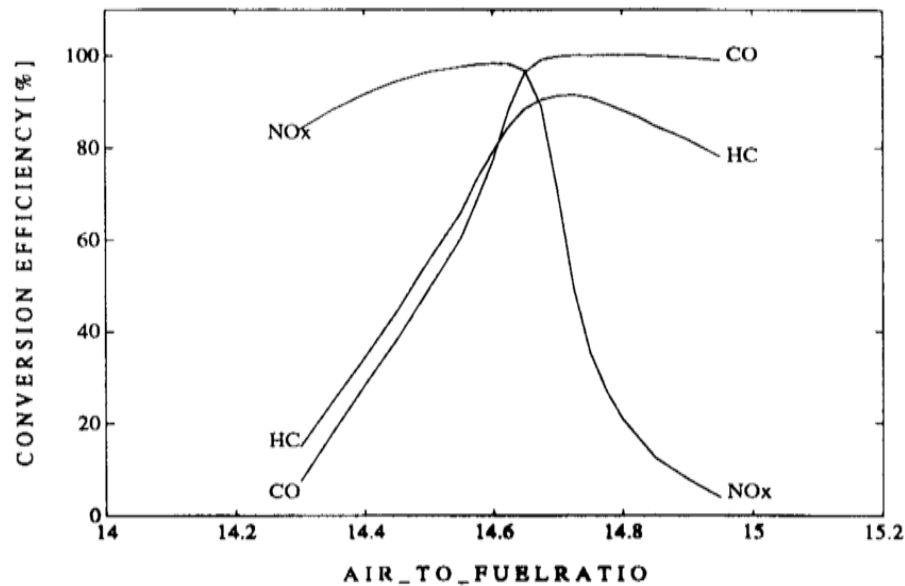
The actuation of the intake valves is decoupled from the camshaft motion.

https://www.youtube.com/watch?v=Td9Gz_h7Qpg



Exhaust aftertreatment systems

Catalytic converter



- A first catalyst is used to oxidize the incomplete combustion products (CO+HC). This catalyst operates correctly in stoichiometric to lean mixture conditions. **This is used in both SI and Diesel engines.**
 - **In Diesel engines**, the system is denominated DOC (Diesel Oxidized Catalyst) and operates at a temperature above 200°C.
 - **In SI engines**, the catalyst is Platinum/Palladium based and has an operating temperature above 600-700°C.
- A second Rhodium Based catalyst is used to reduce the nitrogen oxides. This catalyst operates in rich to stoichiometric conditions. **It cannot be used in Diesel engines for their excess air operation.**
- In order to obtain the best conversion efficiency for all pollutants, the mixture shall be approximately stoichiometric (**lambda sond**).

In order to monitor the catalyst operation and **diagnose faults/degradation**, the **lambda sond** is duplicated. **The two lambda sonds are installed uphill and downhill the catalyst.** **As long as the two lambda sonds detect different oxygen concentrations, the catalyst is operating correctly.** **When the two concentration are too similar, catalyst degradation/fault is detected.**



Exhaust aftertreatment systems

DPF/GPF



DPF

- The system is basically constituted by a metal or ceramic matrix that mechanically captures the soot particles.
- The metal matrix may be loaded or constituted by a catalytic element that lowers the temperature when the oxidation process starts.
- During operation, especially in traffic driving conditions (low load, high EGR, low exhaust temperature), the carbon particles build up inside the filter.
- The pressure drop across the DPF increases progressively as the soot deposits build up. The pressure drop is measured by a **differential pressure sensor**.
- When the pressure drop increases above a certain threshold, the ECM actuates the **DPF regeneration strategy**:
 - Fuel post injection (depending on current load conditions) in order to increase exhaust temperature ($T_{ex} > 450-500^{\circ}\text{C}$).
 - Dedicated low-EGR strategy (increase oxygen availability at the exhaust)

GPF

- The system is similar to DPF.
- Soot formation is relevant for **direct injected (GDI) SI engines only**.
- The pressure drop across GPF is monitored to trigger GPF regeneration.
- In SI engines, the exhaust temperature is higher, so **no additional fuel has to be burned** to raise the exhaust temperature.
- GDI – **SI engines operate with less oxygen excess** respect to Diesel engines. GPF regeneration is executed during cut-offs (deceleration) and/or by increasing airflow (through throttle or VVL) and reducing spark advance (in order to prevent knock).

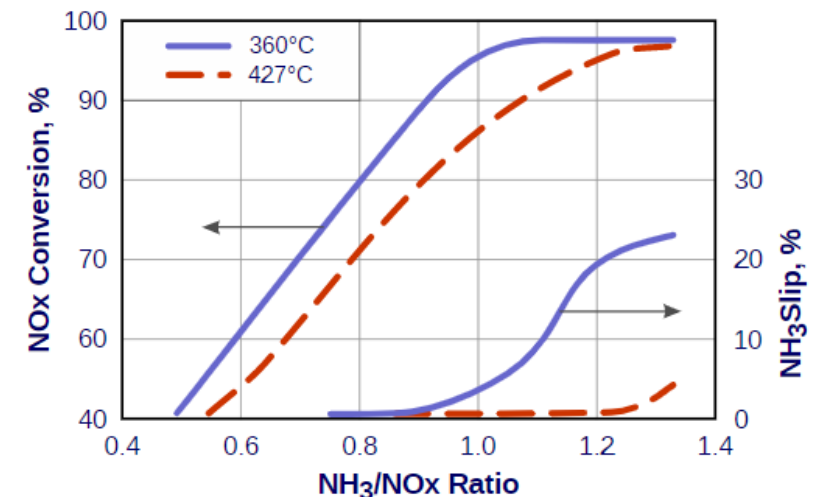


Exhaust aftertreatment systems

SCR (Selective Catalytic Reduction)



- An aqueous solution of urea (AdBlue) is injected uphill a dedicated reactor.
 - The urea decomposes to form **ammonia**, that is the actual reactant.
 - **Ammonia is flammable, toxic and volatile**, that's the reason why urea (safe) is used instead.
 - The reactor is made of multiple channels where the exhaust-ammonia current mixes up and NOx reduction by means of a Vanadium-based catalyst (selective) takes place.
 - The ammonia reduces the NOx so that Nitrogen and Oxygen are the results of the reaction.
- Nitrous oxide concentration is very low in the exhaust gases (1%) so the dwelling time of the exhaust-ammonia current in the catalyst shall be long enough to assure good conversion efficiency.
 - The more ammonia, the higher the NOx reduction rate.
 - Increasing the amount of ammonia increases the ammonia that slips to the exhaust (**to be avoided**).
 - For this reason, the SCR has a second stage ASC (Ammonia slip catalyst) that has the ability to absorb excess ammonia (slipped from the first stage) and use it when the amount of NOx produced increases (works as an ammonia buffer) and the conversion into the first stage is insufficient.





Ducato installation

PACKAGING

- Cylinder Block
- EGR Module
- Turbocharger

DURABILITY

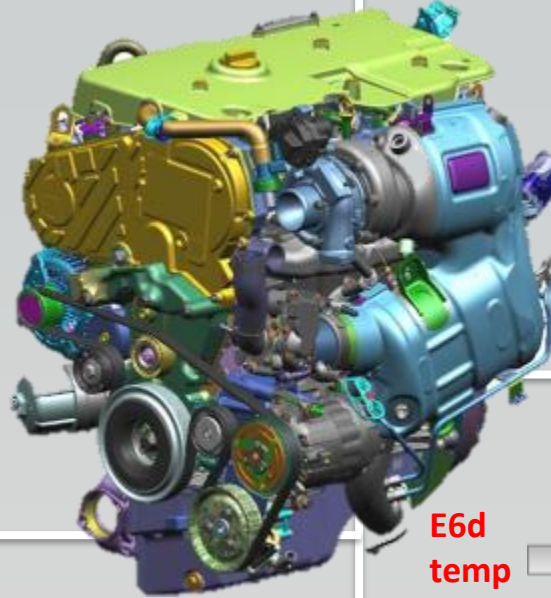
- Cylinder head
- Oil pump G-Rotor
- Oil cooler module
- Water pump
- Steel piston

Compliance

COMBUSTION

- Injectors nozzles
- LP EGR cooled
- Ultrasonic Oil Level sensor (c.o. AR prj)

Reference Engine:
2.2lt Eu6d Temp 200 Hp LIGHT - Cherokee

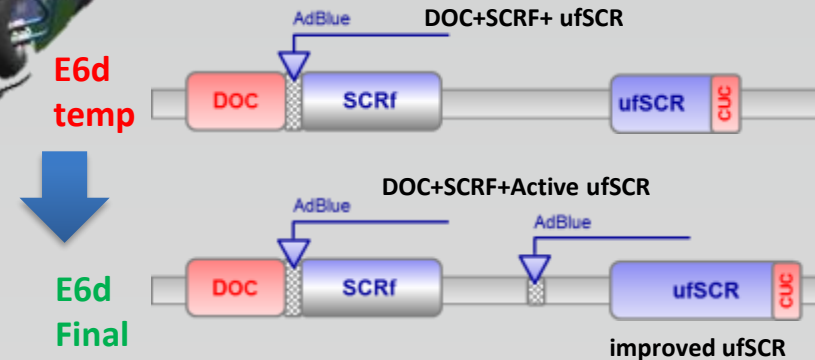


Engine Control Unit (ECU)

- New RB ECU MD1

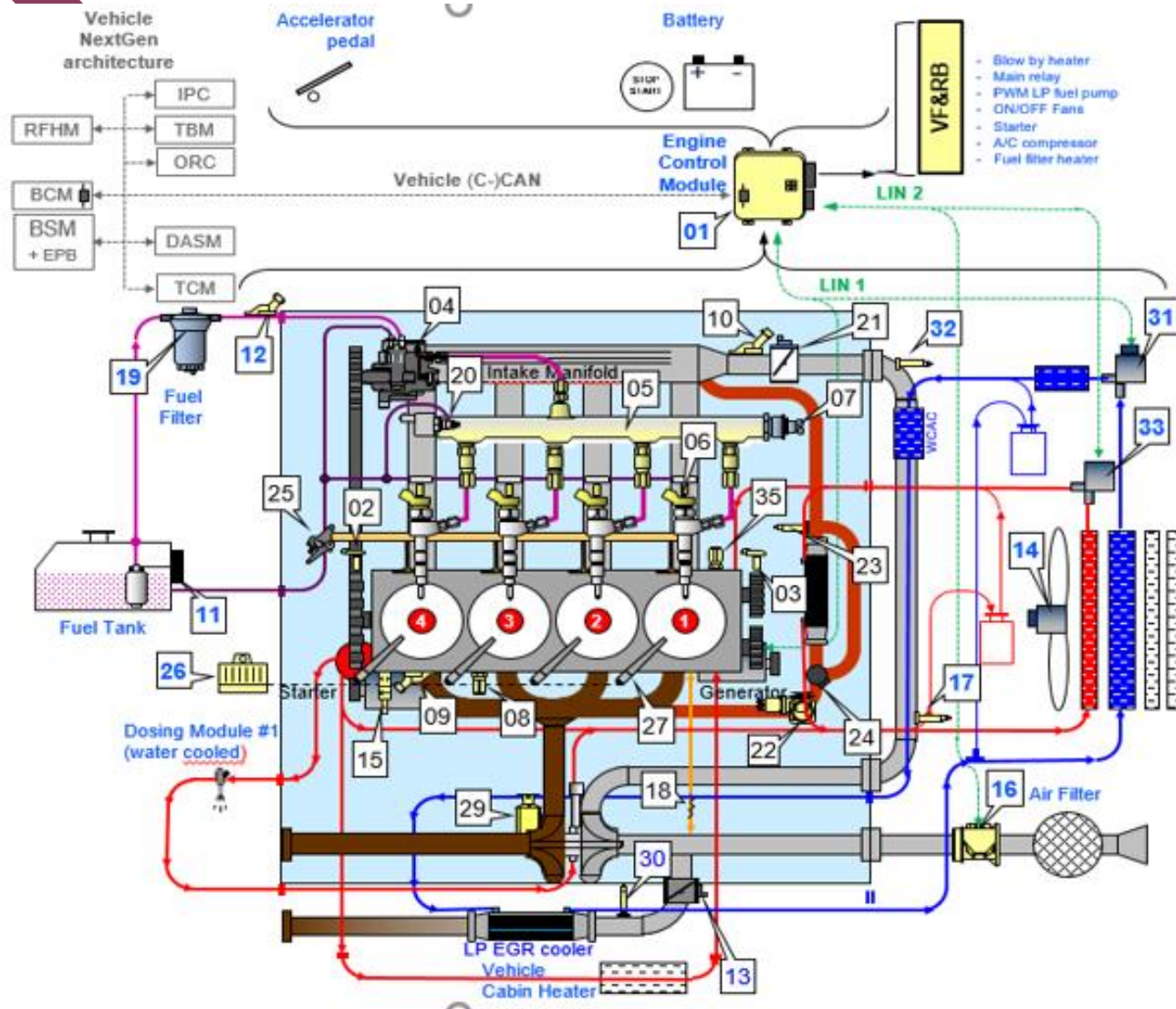
After Treatment System

- 2nd urea injector w/ new Controller (SCU)
- Improved ufSCR
- 3rd Nox sensor
- 2nd water cooling pump (80W)



2.2L Ds Full 180Hp

Engine Components

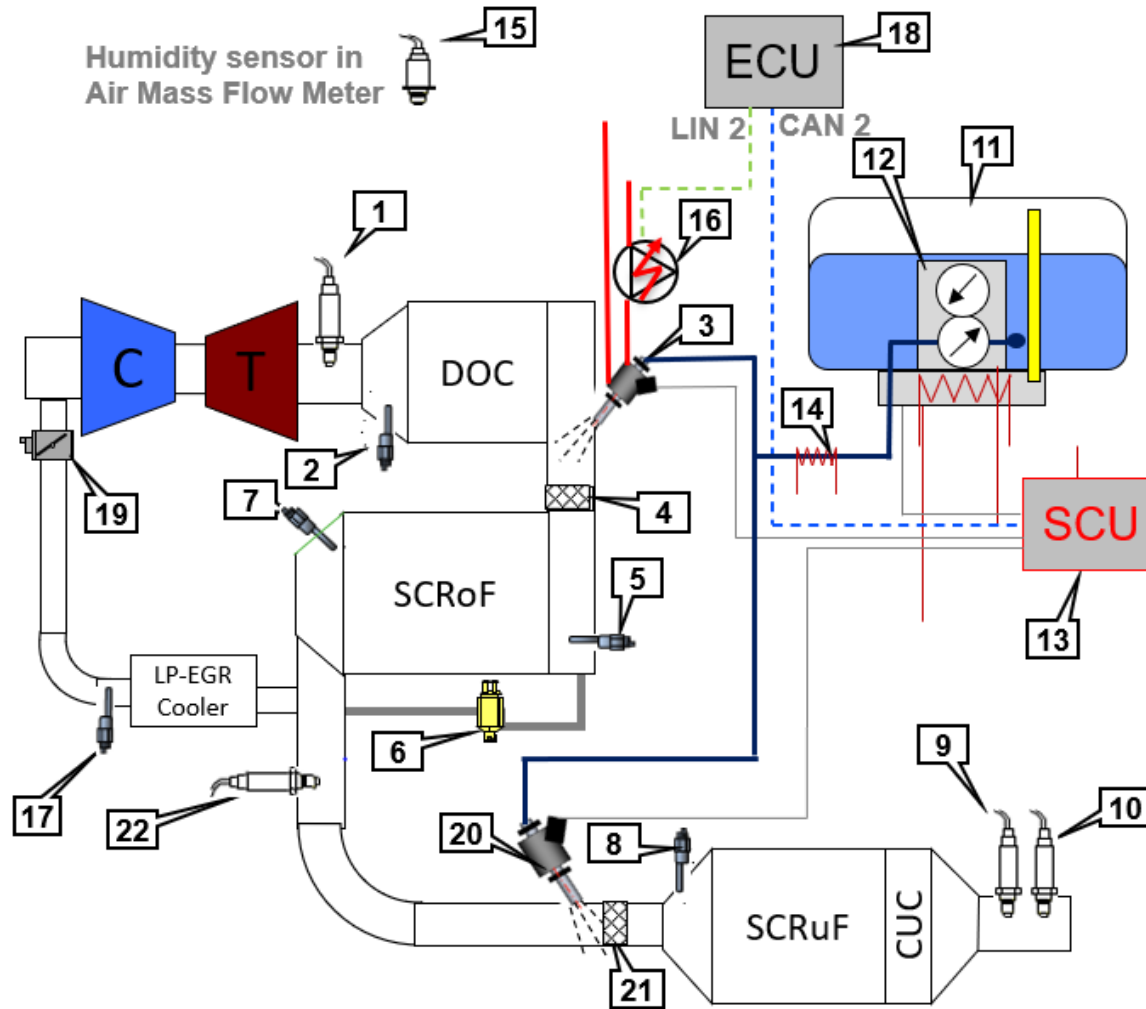


01. Engine Control Module
02. Crankshaft (speed) sensor
03. Camshaft (position) sensor
04. HP fuel pump w/ flow control valve (Metering Unit)
05. HP fuel rail
06. HP fuel injector (4x) CRI 2.20
07. HP fuel pressure sensor
08. Engine coolant temperature sensor
09. Engine oil pressure
10. Intake manifold air pressure & temperature sensor
11. Variable speed LP fuel pump control unit (FPCM) – (PWM fuel supply pump) (OPT)
12. LP fuel pressure sensor (OPT)
13. LP EGR control valve w/ position sensor (H-bridge)
14. Double ON/OFF fans (2 relays)
15. Variable displacement oil pump (mechanical)
16. Air mass meter via LIN with temperature, humidity & pressure sensors
17. outlet compressor air temperature sensor
18. Blow-by heater (1 relay & 1 feedback)
19. Fuel T sensor, water-in-fuel sensor and fuel heater (fuel heater managed by BCM)
20. HP fuel pressure regulator (DRV)
21. Throttle valve actuator (H-bridge)
22. HP EGR control valve w/ position sensor (H-bridge)
23. HP EGR temperature sensor downstream
24. HP EGR cooler bypass actuator
25. Variable smart swirl actuator
26. Glow Plugs Control Unit (GCU3)
27. Glow plug (4x)
29. Electrical VGT valve w/ position sensor (H-bridge)
30. LP EGR temperature sensor
31. Secondary circuit cooling pump via LIN1 (80 W new)
32. Outlet WCAC air temperature sensor
33. Primary circuit cooling pump via LIN2 (20W carry over Ducato NP)
35. Ultrasonic oil level sensor



SCRoF + SCRuF + double dosing modules

ATS System



* Humidity sensor could be used to estimate condensation in LP-EGR Cooler

1. DOC inlet NOx Sensor
2. DOC inlet temperature sensor (T4)
3. Urea Dosing Module (water cooled)
4. Urea mixer
5. SCRoF inlet temperature sensor (T5)
6. SCRoF diff. Pressure sensor
7. SCRoF outlet temperature sensor (T6)
8. SCRuF inlet temperature sensor (T7)
9. Soot sensor (Bosch sensor)
10. SCRuF outlet NOx sensor
11. Urea tank assembly
12. Urea pump module w/ temperature, level, quality sensor and tank heater
13. SCR Control Unit
14. Urea heated pipe
15. Humidity sensor (in air flow meter)*
16. Primary circuit cooling pump (#33 in slide 2)
17. LP-EGR out temperature sensor
18. ECU
19. LP EGR Valve
20. Urea Dosing Module (Air cooled)
21. Underfloor Mixer
22. SCRoF outlet NOx sensor

CONTACTS

Fabrizio Donatantonio

Fabrizio.donatantonio@kineton.it

Via E. Gianturco, 23
(presso complesso Mecfond)
80146 Napoli

www.kineton.it

